

Product Walkthrough: Origin Studios

A plain-English walkthrough of the current codebase for a non-technical stakeholder.

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Security note

This document intentionally excludes pasted secrets, database passwords, service-role keys, and private environment values.

Overview

Origin Studios is a creative platform for planning, generating, organizing, and editing visual scenes. The active product today is Origin Frame, which combines scene-based project creation, AI-assisted Shot Design, media storage, and a browser video editor.

The roles and personas found in the code are: public visitor, registered user, workspace owner, workspace admin, workspace editor, workspace viewer, project collaborator editor, project collaborator viewer, and global admin. The global admin flag exists but does not currently drive a visible product flow.

Feature-by-feature breakdown

Public landing page and studio catalog

What it is for	This introduces Origin Studios and lets a visitor choose a product room.
Who can access it	Anyone.
User journey	The visitor enters the studio catalog, sees Origin Frame as active, and sees Origin Sites and Origin Garage as coming soon.
Conditional logic and edge cases	Sites and Garage are disabled placeholders.

Authentication and account access

What it is for	This lets people create accounts, sign in, sign out, and recover passwords without Supabase Auth.
Who can access it	Public visitors and signed-in users.
User journey	A user signs up with name, email, and password. The system creates the account, stores a protected password hash, and creates a personal workspace. Login creates a secure session. Password reset sends or logs a reset link and revokes old sessions after reset.
Conditional logic and edge cases	Signup currently marks accounts verified immediately, while the UI still mentions inbox verification.

Origin Frame app shell

What it is for	This is the main logged-in workspace shell.
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Who can access it	Any signed-in user.
User journey	The user enters Origin Frame and sees navigation for Projects, Video editor, Costs, Settings, Sign out, and identity.
Conditional logic and edge cases	Settings is visible but not a full screen yet. Costs data is owner-only.

Workspaces and roles

What it is for	Workspaces group people, projects, media, costs, and permissions.
Who can access it	Signed-in users; owners and admins have management access.
User journey	Signup creates a personal workspace and owner membership. Users can also create team workspaces. Backend routes support members, roles, removals, invitations, and invite acceptance.
Conditional logic and edge cases	Member and invite management are backend-ready but lack full frontend UI.

Project dashboard

What it is for	This is the project shelf for scene-based creative projects.
Who can access it	Viewers can see projects. Editors can create projects.
User journey	The dashboard loads accessible workspaces and projects. The user searches projects, creates a new project, chooses a workspace, adds a name and note, then opens the project.
Conditional logic and edge cases	Project cards may show zero scenes because list data does not include full scene details.

Project workspace

What it is for	This is where a project is built as ordered scenes.
Who can access it	Viewers can inspect. Editors can create and edit.
User journey	Opening a project loads scenes into a workspace with top bar, left rail, center stage, right history, and bottom filmstrip.
Conditional logic and edge cases	Library, Sequence, Settings, Collaborators, Comments, and Preview are present but not fully complete flows.

Scene creation and cards

What it is for	Scenes are ordered building blocks.
Who can access it	Editors create; viewers inspect.
User journey	The user adds a scene, chooses a starting format, and adds a note. The system creates the scene and its initial version.
Conditional logic and edge cases	Video scenes show play; image scenes do not.

Scene media upload

What it is for	This attaches real media to a scene.
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Who can access it	Project editors.
User journey	The browser requests a signed upload URL, uploads to Supabase Storage, confirms the upload, and creates a scene version pointing to the new asset.
Conditional logic and edge cases	Only approved file types are accepted. Missing storage config causes upload failure.

Scene history and versions

What it is for	This preserves how a scene changes over time.
Who can access it	Viewers can see history. Editors create versions.
User journey	Uploads and generated outputs create new immutable scene versions, shown in the right history panel.
Conditional logic and edge cases	Make-current exists in the backend but not in the UI.

Scene ordering and filmstrip

What it is for	This lets users arrange scene order.
Who can access it	Editors reorder; viewers inspect.
User journey	The user drags a filmstrip handle. The frontend updates immediately and the backend saves the new order if the project revision still matches.
Conditional logic and edge cases	Revision conflicts reload canonical state but do not show a detailed conflict resolver.

Shot Design workflows

What it is for	This is the AI generation area.
Who can access it	Project editors.
User journey	The user picks one of seven workflows, enters prompt/settings/provider/model, and starts a job. A worker contacts the provider and returns outputs that can be attached to a scene.
Conditional logic and edge cases	All modes share a generic form. Provider models are seeded disabled and require setup.

Provider jobs and worker

What it is for	This runs long AI and render jobs outside page requests.
Who can access it	Triggered indirectly by editors; operated by the backend worker.
User journey	Jobs move from queued to running to succeeded or failed. Successful generation can record cost events.
Conditional logic and edge cases	Missing credentials fail jobs clearly. Polling has a bounded window.

Assets and storage

What it is for	Assets store media used by scenes, generated outputs, and edits.
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Who can access it	Workspace viewers can list. Editors can create versions/delete.
User journey	Uploads and generation promotion create assets and asset versions stored in Supabase Storage.
Conditional logic and edge cases	No full frontend asset library. Collections exist in the database but not the UI.

Live collaboration

What it is for	This keeps project screens roughly synchronized.
Who can access it	Authenticated project viewers.
User journey	The frontend joins a project room. Scene creation, reorder, and version events cause other clients to reload.
Conditional logic and edge cases	Presence exists but is not displayed. Comments are backend-only for now.

Comments and mentions

What it is for	This supports feedback attached to project objects.
Who can access it	Project viewers according to current backend rules.
User journey	Comments can be created through the API and broadcast over realtime.
Conditional logic and edge cases	No frontend comment UI. Viewer resolve permission may need product review.

Collaborators and share links

What it is for	These extend access beyond workspace membership.
Who can access it	Project admins.
User journey	Admins can add existing users as project collaborators or create public share tokens.
Conditional logic and edge cases	No frontend UI and public shares only return limited metadata.

Video project library

What it is for	This is the entry point for timeline editing.
Who can access it	Workspace viewers can list. Editors create.
User journey	The user opens Video editor, creates an untitled edit in the first workspace, and opens it.
Conditional logic and edge cases	No workspace picker. Scenes are not yet sent directly into a timeline.

Browser video editor

What it is for	This assembles media into a timeline and previews it in the browser.
Who can access it	Viewers can open. Editors can import, edit, and save.

User journey	Users import media, preview with Remotion, drag the playhead, trim, split, move clips, add captions, adjust volume/mute, undo/redo, autosave, and save.
Conditional logic and edge cases	Export is disabled. Empty timelines start at 30 seconds; real timelines follow the last clip end.

Frame-level video instructions

What it is for	This lets users select exact frames and save edit instructions.
Who can access it	Video project editors.
User journey	The browser estimates FPS, calculates frame count, creates thumbnails, and lets the user select a frame and save an instruction.
Conditional logic and edge cases	Instructions are metadata only and do not yet alter video pixels.

Remotion preview and future render path

What it is for	This previews timelines now and prepares future export.
Who can access it	Video editor users for preview; editors for backend render jobs.
User journey	Remotion renders video, images, audio, captions, and black gaps from the timeline document.
Conditional logic and edge cases	Frontend export is disabled. Backend render needs AWS/Remotion configuration.

Cost observatory

What it is for	This shows observed provider costs.
Who can access it	Workspace owners.
User journey	Owners open Costs and see provider summaries and recent events.
Conditional logic and edge cases	This is not billing. There are no credits, subscriptions, budgets, or payment flows.

Operational endpoints and clean-room safeguards

What it is for	These support deployment, docs, and source hygiene.
Who can access it	Operators and developers.
User journey	Health/readiness endpoints report service state. Clean-room scans fail on forbidden legacy identifiers or dependencies.
Conditional logic and edge cases	OpenAPI is hand-authored and not a full source of truth.

Cross-feature flows

- Signup to first project: signup creates a user and personal workspace, then the user creates and opens a project.
- Scene upload to version history: upload goes to Supabase Storage, becomes an asset, then becomes a new scene version.
- Shot Design to scene media: prompt creates a job, provider output is imported, and a result is promoted to a selected scene.
- Scene ordering to realtime updates: reorder saves a sequence and broadcasts reload events to connected clients.
- Video import to saved timeline: imported media becomes a timeline item, can be edited, and is saved as a timeline version.
- Generation to costs: successful generation records a cost event visible to workspace owners.
- Workspace invitation: an invited email accepts a token and joins the workspace with the assigned role.

What is fully working or substantially wired

- Landing page and catalog.
- Custom signup, login, logout, and password reset.
- Secure sessions and CSRF handling.
- Personal workspace creation.
- Project listing and creation.
- Scene creation, uploads, versions, history, and reorder.
- Basic realtime project invalidation.
- Shot Design job creation and polling.
- Provider worker architecture.
- Generation output promotion to scenes.
- Supabase Storage upload and confirmation.
- Video project listing and creation.
- Browser video editor import, preview, trim, split, captions, frame selection, undo/redo, autosave, and save.
- Owner-only cost observatory.
- Clean-room source scan.

What is stubbed, incomplete, or partially wired

- Origin Sites and Origin Garage are placeholders.
- Email verification UI conflicts with currently bypassed backend verification.
- Workspace members, invitations, collaborators, comments, share links, and presence need frontend UI.
- Project Library, Sequence, Settings, Collaborators, Comments, and Preview are not complete flows.
- Shot Design modes need dedicated forms and reference selection.
- Provider models are disabled by default and need admin/configuration UX.
- Video export is disabled in the frontend.
- Backend render jobs need AWS/Remotion Lambda setup.
- Frame instructions are saved but not applied to video pixels.
- Timeline transitions and transform editing are not fully exposed.

- Collections, activity events, and global admin exist but are not active product flows.

Open questions

- Should email verification remain disabled, or should the signup UI change?
- Should viewers be able to resolve comments?
- What should public share-link roles allow?
- Should project collaborators require existing accounts?
- What belongs in Project Settings first?
- Should each Shot Design mode get a dedicated flow?
- Who enables provider models and credentials in production?
- Should timeline FPS follow source FPS, output FPS, or both?
- Should frame thumbnails be persisted as sprites?
- Should frame instructions apply only to one frame or become keyframes over a range?
- Should export use Remotion Lambda only, local worker fallback, or both?
- Should project scenes be inserted directly into a video timeline?
- What is the intended use of global admin?